ST03

Biology: Cell, Tissues and the Human Body

Science and Technology is 18-23 questions in every RAS paper and biology feeds 5-7 of them. This chapter is the single biggest supplier. Blood groups and the Rh factor are a documented repeat - 2013, 2021 and 2023 - so that one section alone is worth more than the rest of the chapter put together.

This is the chapter about you - the cell you are built from and the nine systems that keep you running. RPSC does not ask biology as reasoning. It asks it as labels: which organelle, which enzyme, which gland, which blood group. So the work here is naming, not theorising, and every label you fix is a mark banked.

Cell, Tissues and the Human Body

The cell
- Hooke 1665, cork
- Schleiden + Schwann + Virchow
- Mitochondrion - powerhouse
- Lysosome - suicide bag
- Plant cell vs animal cell

Cell division
- Mitosis - 2 diploid, somatic
- Meiosis - 4 haploid, gametes
- Crossing over at pachytene
- PMAT sequence

Plant side
- Whittaker 1969, five kingdoms
- Photosynthesis - grana + stroma
- O2 from photolysis of water
- Auxin, gibberellin, ethylene

Digestion + breathing
- Ptyalin, pepsin, trypsin
- Liver - largest gland
- Absorption in small intestine
- 12 pairs of ribs, tidal volume 500 ml

Heart and blood
- 4 chambers, SA node pacemaker
- 120/80 mm Hg, 72 per minute
- RBC 120 days, spleen graveyard
- Harvey - circulation

Blood groups (repeat)
- Landsteiner 1901
- O universal donor, AB recipient
- Rh from Rhesus monkey, 1940
- Erythroblastosis foetalis, anti-D

Kidney and nerves
- Nephron, 10 lakh per kidney
- Ureotelic, urea in liver
- Cerebrum largest, cerebellum balance
- 12 cranial + 31 spinal nerves

Hormones and bones
- Pituitary - master gland
- Insulin - beta cells
- 206 bones, femur longest, stapes smallest
- 46 chromosomes, XX and XY

The cell – theory, the men who named it, and the organelle table

Start with who said what. Robert Hooke looked at a thin slice of cork in 1665 and called the empty boxes he saw 'cells'. Those were dead cell walls. Anton van Leeuwenhoek was the first to see living cells. Robert Brown found the nucleus in 1831. Then came the cell theory. Learn the three names in order, because RPSC asks the third one most often.

- Cell theory - M.J. Schleiden (1838, plants) and Theodor Schwann (1839, animals).
- Rudolf Virchow (1855) added 'Omnis cellula e cellula' - every cell arises from a pre-existing cell.
- Prokaryote - no true nucleus, no membrane-bound organelles, 70S ribosome (bacteria, blue-green algae).
- Eukaryote - true nucleus with a membrane, organelles present, 80S ribosome in the cytoplasm.
- Plasma membrane is selectively permeable - the fluid mosaic model of Singer and Nicolson (1972).
- The largest human cell is the ovum; the longest is the nerve cell (neuron).

Scientist to discovery - a ready-made match-the-following

Scientist	Discovery / contribution
Robert Hooke	Coined the term 'cell', 1665, from cork
Anton van Leeuwenhoek	First saw living cells
Robert Brown	Nucleus of the cell, 1831
Schleiden and Schwann	Cell theory, 1838-39
Rudolf Virchow	Omnis cellula e cellula, 1855
Camillo Golgi	Golgi apparatus
Christian de Duve	Lysosome
Watson and Crick	Double helix structure of DNA, 1953

Organelle, function and the nickname RPSC hides the question behind

Organelle	Chief function	Nickname / key point
Nucleus	Controls the cell, holds the DNA	Discovered by Robert Brown, 1831
Mitochondrion	ATP through the Krebs cycle and oxidative phosphorylation	Powerhouse of the cell; double membrane, inner one folded into cristae; own circular DNA and 70S ribosomes

Organelle	Chief function	Nickname / key point
Ribosome	Protein synthesis	70S in prokaryotes (50S + 30S), 80S in eukaryotes (60S + 40S)
Endoplasmic reticulum	Rough ER - protein transport; smooth ER - lipid synthesis	The transport channel of the cell
Golgi apparatus	Packaging, secretion, and formation of lysosomes	Named after Camillo Golgi
Lysosome	Intracellular digestion by hydrolytic enzymes at acidic pH	Suicide bag, named by Christian de Duve
Chloroplast	Photosynthesis - light reaction in grana, Calvin cycle in stroma	Kitchen of the cell; chlorophyll holds magnesium
Chromoplast	Gives colour to flowers and fruits	A plastid; plant cells only
Leucoplast	Storage of starch, oil and protein	A colourless plastid
Vacuole	Storage and turgidity	Large and central in the plant cell
Centriole / centrosome	Forms the spindle during cell division	Animal cells; absent in higher plants
Cell wall	Rigidity and protection; made of cellulose	Plants only - fungi have chitin, bacteria peptidoglycan

Plant cell vs animal cell - the exact contrast RPSC sets

Plant cell	Animal cell
Cellulose cell wall outside the membrane	No cell wall, only the plasma membrane
Plastids present - chloroplast, chromoplast, leucoplast	Plastids absent
One large central vacuole	Vacuoles small, many or absent
Centriole absent in higher plants	Centriole and centrosome present
Lysosomes rare	Lysosomes prominent
Reserve food is starch	Reserve food is glycogen
Cytokinesis by a cell plate	Cytokinesis by furrowing of the membrane

TRAP

Two mismatches RPSC keeps planting. First - 'Ribosome: synthesis of lipids' is WRONG; ribosomes make protein, the smooth ER makes lipid. Second - do not say the centriole is absent from all plants; say it is absent from higher plants. Also note chlorophyll contains magnesium while haemoglobin contains iron - that pair is a favourite one-liner.

Cell division, biomolecules and the nucleic acids

Two kinds of division, two entirely different purposes. Mitosis keeps you growing and repairs your wounds - it copies a cell exactly. Meiosis makes eggs and sperm - it halves the chromosome number and shuffles the genes. If you fix that purpose, the rest of the table writes itself.

The sequence of mitosis - asked as an ordering question

Interphase (DNA copied) → Prophase → Metaphase (chromosomes line up at the equator) → Anaphase (sister chromatids separate) → Telophase → Cytokinesis

Mitosis vs meiosis

Mitosis	**Meiosis**
Takes place in somatic (body) cells	Takes place in germ cells of the reproductive organs
One division, two daughter cells	Two divisions, four daughter cells
Daughter cells diploid (2n) and identical to the parent	Daughter cells haploid (n)
Equational division - chromosome number unchanged	Reductional division - chromosome number halved
No crossing over, so no variation	Crossing over at pachytene of prophase-I creates variation
For growth, repair and healing	For gamete formation - the basis of sexual reproduction

- Crossing over happens in the pachytene sub-stage of prophase-I; the chiasmata become visible later, at diplotene.
- Prophase-I sub-stages in order - leptotene, zygotene, pachytene, diplotene, diakinesis.
- Humans have 46 chromosomes (23 pairs); a gamete carries 23.
- Sex determination - XX is female, XY is male; the sperm decides the sex of the child.

TRAP

'Meiosis takes place in the somatic cells of the body' is a standard false statement in RAS statement-based questions. Meiosis is germ-cell only. And remember mitosis is equational, meiosis-I is the reductional one - not meiosis-II.

Four families of molecules build the body - carbohydrate, protein, fat and nucleic acid - and enzymes run the reactions between them. RPSC asks this at a very basic level: energy value per gram, what an enzyme does, and the difference between DNA and RNA. Nothing deeper.

- Energy value - carbohydrate 4 kcal/g, protein 4 kcal/g, fat 9 kcal/g. Fat is more than double, and that is the asked fact.
- Carbohydrate is the immediate fuel; protein builds and repairs; fat stores energy and insulates.
- Enzymes are proteins - except ribozymes, which are RNA. They lower the activation energy of a reaction.
- Enzymes are specific, are not consumed in the reaction, and are sensitive to temperature and pH.
- DNA double helix - James Watson and Francis Crick, 1953, using Rosalind Franklin's X-ray work.
- Base pairing - adenine with thymine by two hydrogen bonds, guanine with cytosine by three.
- So G-C rich DNA needs more heat to melt - that is why the number of bonds gets asked.
- RNA has uracil in place of thymine, and ribose sugar in place of deoxyribose.
- DNA is usually double-stranded and lives in the nucleus; RNA is single-stranded and works in the cytoplasm.
- Memory line - 'AT the Two, GC the Three': Adenine-Thymine two bonds, Guanine-Cytosine three.

Classification, tissues and how a plant works

Two names carry this whole area. Carolus Linnaeus gave binomial nomenclature - every organism gets a two-word Latin name, genus first with a capital, species second in lower case, the whole thing italicised or underlined. R.H. Whittaker in

1969 divided life into five kingdoms. Learn the year and the five names; that is the question.

- Five kingdoms (Whittaker, 1969) - Monera, Protista, Fungi, Plantae, Animalia.
- Basis of the division - cell structure, body organisation, mode of nutrition and reproduction.
- Monera are prokaryotes (bacteria, cyanobacteria); the rest are eukaryotes.
- Binomial nomenclature - Homo sapiens: Homo is the genus, sapiens the species.
- Plant tissues - meristematic (dividing, at growing tips) and permanent (parenchyma, collenchyma, sclerenchyma; xylem carries water, phloem carries food).
- Animal tissues - four types: epithelial (covering), connective (blood, bone, cartilage), muscular and nervous.

Photosynthesis, step by step

Light falls on chlorophyll in the grana → Photolysis of water - oxygen is released here →

ATP and NADPH formed (light reaction, in the grana) → Calvin cycle fixes CO_2 (dark reaction, in the stroma) → Glucose

- Overall equation - $6CO_2 + 6H_2O$ gives $C_6H_{12}O_6 + 6O_2$, in sunlight and chlorophyll.
- The oxygen released comes from WATER, not from carbon dioxide - proved with heavy-oxygen labelled water.
- Light reaction in the grana or thylakoid; Calvin or dark reaction in the stroma.
- Chlorophyll contains magnesium at its centre; haemoglobin contains iron.
- Transpiration is loss of water as vapour, chiefly through stomata - it pulls water up the plant.

Plant hormones and their principal effect

Hormone	Principal effect
Auxin	Apical dominance, cell elongation, phototropism
Gibberellin	Stem elongation, bolting in rosette plants, seed germination
Cytokinin	Cell division, delays senescence (ageing) of leaves
Abscisic acid	The 'stress hormone' - closes stomata, promotes dormancy
Ethylene	The only gaseous hormone - ripening of fruit and senescence

The digestive system - enzyme, source, substrate

Digestion is a relay of enzymes, and the exam question is always 'which enzyme, from where, acting on what'. Starch is attacked first in the mouth, protein in the stomach, and everything is finished in the small intestine, where absorption also happens through the villi. Bile is the trick answer - it is a secretion, not an enzyme.

The path of swallowed food

Mouth → Oesophagus → Stomach → Duodenum → Jejunum → Ileum → Large intestine → Rectum and anus

Digestive secretions - source and action

Secretion	Source	Acts on
Ptyalin (salivary amylase)	Salivary glands	Starch to maltose, in the mouth
Pepsin (secreted as pepsinogen)	Gastric glands, needs HCl	Protein to peptones, in the stomach
Rennin	Gastric glands of infants	Curdles the casein of milk
Bile - contains NO enzyme	Liver, stored in the gall bladder	Emulsifies fat, makes the medium alkaline

Secretion	Source	Acts on
Trypsin	Pancreas (as trypsinogen)	Protein and peptones to peptides
Pancreatic amylase and lipase	Pancreas	Starch to maltose; fat to fatty acids and glycerol
Maltase, sucrase, lactase, peptidases	Intestinal juice	Final breakdown before absorption

- Liver is the largest gland in the body, about 1.5 kg - it makes bile, forms urea, stores glycogen, iron and fat-soluble vitamins, and detoxifies.
- The liver does NOT make insulin. That is the pancreas.
- Pancreas is a mixed or heterocrine gland - exocrine (digestive juice) plus endocrine (insulin, glucagon).
- Maximum absorption of food takes place in the small intestine, through finger-like villi that multiply the surface area.
- The appendix is a vestigial organ in humans; the largest and strongest muscle-lined organ of chemical digestion is the stomach.
- HCl of the stomach kills bacteria and gives pepsin the acidic pH it needs.
- Memory line - the three P's in canal order: Ptyalin (mouth), Pepsin (stomach), Pancreas (small intestine).

The respiratory system

The path of air, nose to alveolus

Nostrils → Pharynx → Larynx (voice box) → Trachea →

Bronchi → Bronchioles → Alveoli - the actual site of gas exchange

Breathing is mechanics; respiration is chemistry. Air enters through the nose, passes the pharynx, larynx (the voice box), trachea and bronchi, and reaches the alveoli of the lungs, where the actual gas exchange happens. The muscles that do the pumping are the diaphragm and the intercostals between the ribs.

- Twelve pairs of ribs form the rib cage; the diaphragm plus the intercostal muscles drive breathing.
- Tidal volume - the air moved in one quiet breath - is about 500 ml.
- Normal breathing rate is about 12-16 per minute at rest.
- Oxygen is carried in the blood mainly as oxyhaemoglobin, bound to the iron of haemoglobin in the RBC.
- Carbon dioxide is carried mostly as bicarbonate dissolved in the plasma, not on haemoglobin.
- Alveoli are the site of exchange; the right lung has three lobes, the left has two.
- The medulla oblongata, not the lungs, sets the rhythm of breathing.
- Do not write 'CO2 is carried as carboxyhaemoglobin' - carboxyhaemoglobin is haemoglobin bound to carbon MONOXIDE, the poison.

The circulatory system - heart, blood and the pump

The human heart has four chambers - two atria on top, two ventricles below - and is made of cardiac muscle, which is striated but involuntary and never tires. It beats on its own because of the sino-atrial node, a patch of tissue in the RIGHT atrium that fires the impulse. That word 'right' is the whole question.

- SA node in the right atrium is the pacemaker; the AV node lies in the inter-atrial septum.
- 'Lub' - the first heart sound - is the closure of the atrio-ventricular valves. 'Dub' is the closure of the semilunar valves.
- Normal blood pressure 120/80 mm Hg; 120 is systolic (contraction), 80 diastolic (relaxation).

- Resting heart rate about 72 beats per minute.
- William Harvey described the circulation of blood, in 1628.
- Blood pressure is measured with a sphygmomanometer; the ECG records the electrical activity of the heart.
- Humans have a closed, double circulation - pulmonary (heart to lungs) and systemic (heart to body).
- Arteries carry blood away from the heart (oxygenated, except the pulmonary artery); veins bring it back (deoxygenated, except the pulmonary vein).

What blood is made of

Component	Normal value	Key point
Plasma	About 55% of blood volume	90-92% water; carries most of the CO2 as bicarbonate
RBC (erythrocyte)	4.5-5.5 million per microlitre; lives 120 days	No nucleus in mammals; made in red bone marrow, destroyed in the spleen
WBC (leucocyte)	About 5,000-10,000 per microlitre	Defence; neutrophils are the most numerous, lymphocytes make antibodies
Platelets (thrombocyte)	1.5-4.5 lakh per microlitre	Blood clotting

TRAP

'Mature mammalian red blood corpuscles possess a nucleus' is FALSE and RPSC uses it. Mammalian RBCs lose the nucleus, which leaves more room for haemoglobin; camels and llamas are the odd exceptions with oval RBCs. The spleen is the 'graveyard of RBCs'.

Blood groups and the Rh factor - the documented repeat

Give this section extra time. RPSC has asked blood groups and the Rh factor in 2013, in 2021 and again in 2023 - three papers out of the last four. The logic is simple once you see it. Your red cell carries an antigen; your plasma carries the antibody against the antigen you do NOT have. A transfusion fails when donor antigen meets recipient antibody and the cells clump.

- The ABO system was discovered by Karl Landsteiner in 1901; he received the Nobel Prize for it in 1930.
- Group A has antigen A on the red cell and anti-B antibody in the plasma. Group B is the mirror image.
- Group AB has both antigens and NO antibody - so it can receive from everyone: the universal recipient.
- Group O has no antigen and both antibodies - so its cells offend nobody: the universal donor.
- Strictly, O-negative is the true universal donor and AB-positive the true universal recipient, once Rh is counted.

ABO system - antigen, antibody and who you can give blood to

Group	Antigen on RBC	Antibody in plasma	Can donate to
A	A	anti-B	A and AB
B	B	anti-A	B and AB
AB	A and B	None	AB only
O	None	anti-A and anti-B	A, B, AB and O - universal donor

The same table read the other way - who you can receive from

Group	Can receive from	Tag
AB	A, B, AB and O	Universal recipient
A	A and O	-
B	B and O	-
O	O only	Universal donor but the fussiest recipient

How ABO is inherited - the genetics behind the group

- ABO is controlled by three alleles of one gene on chromosome 9 - I-A, I-B and i.
- I-A and I-B are codominant with each other; i is recessive to both.
- Genotypes - A is I-A I-A or I-A i; B is I-B I-B or I-B i; AB is I-A I-B; O is ii.
- So an A father and a B mother can produce a child of any of the four groups, including O.
- A child of group O needs both parents to carry the i allele - this is the classic parentage question.
- The rare Bombay blood group (hh), first reported from Mumbai in 1952, types as O but cannot receive ordinary O blood.

ASKED IN RAS

Asked in RAS Pre 2023 - which blood group is the universal donor? Answer: O. The same paper asked the statement set on ABO: AB plasma has neither anti-A nor anti-B, O plasma has both, and Landsteiner discovered the system - all three correct.

ASKED IN RAS

Asked in RAS Pre 2013 as a NOT-matched question - 'Blood group AB : anti-A and anti-B antibodies' is the wrong pair. AB plasma has no antibody at all; that is precisely why AB is the universal recipient.

Now the Rh factor. It is a second, separate antigen on the red cell, found by Landsteiner and A.S. Wiener in 1940 in the Rhesus monkey, Macaca mulatta - hence the name. If you have it you are Rh-positive; about 93-95 per cent of

Indians are. If you do not, you are Rh-negative, and your body will manufacture anti-Rh antibodies the first time it meets Rh-positive blood.

- First exposure sensitises an Rh-negative person; the SECOND exposure is the one that destroys the red cells.
- So an Rh-negative person cannot safely receive Rh-positive blood 'repeatedly' - that statement is always false.
- Rh incompatibility in pregnancy - an Rh-negative mother carrying an Rh-positive foetus.
- The first pregnancy is usually safe; the danger comes from the second pregnancy onwards.
- The condition in the baby is erythroblastosis foetalis, or haemolytic disease of the newborn.
- Prevention - anti-D immunoglobulin given to the mother, usually within 72 hours of delivery.

ASKED IN RAS

Asked in RAS Pre 2021 - the Rh factor is named after the Rhesus monkey. Asked again in RAS Pre 2023 as statements: erythroblastosis foetalis can occur when an Rh-negative mother carries an Rh-positive foetus (true), anti-D prevents it (true), an Rh-negative person can repeatedly receive Rh-positive blood (false).

YAAD RAKHO

Yaad rakho - 'O gives to all, AB takes from all'. O = zerO antigen, so nothing to object to. AB = All antiBodies absent, so nothing to object with. Rh = Rhesus monkey, 1940, Landsteiner and Wiener - and Landsteiner's name therefore attaches to BOTH discoveries.

The excretory system

How urine is made

Blood enters the glomerulus → Ultrafiltration into Bowman's capsule → Selective reabsorption along the tubule → Tubular secretion → Urine collects → Ureter, urinary bladder, urethra

The kidney filters the blood and the nephron does the filtering. Each kidney holds about ten lakh nephrons. Blood enters a knot of capillaries called the glomerulus, sitting inside a cup called Bowman's capsule - together they make the Malpighian corpuscle, where filtration happens. What the body still wants is reabsorbed further down the tubule; what is left is urine.

- Nephron is the structural and functional unit of the kidney - about 10 lakh per kidney.
- Bowman's capsule plus glomerulus = Malpighian corpuscle, the filtration unit.
- Humans are ureotelic - they excrete urea. Birds and reptiles are uricotelic (uric acid); most fish are ammonotelic.
- Urea is made in the LIVER by the ornithine (urea) cycle and only excreted by the kidney.
- ADH or vasopressin, from the posterior pituitary, controls water reabsorption - less ADH means more, dilute urine.
- Urine is about 95 per cent water; the skin and lungs are accessory excretory organs.
- Dialysis is the artificial substitute when the kidneys fail.

- Planted false statement to watch - 'human beings are uricotelic'. We are UREOtelic; birds are uricotelic.

The nervous system, the brain and the sense organs

The nervous system is made of neurons, the longest cells in the body, and it splits into the central system (brain and spinal cord) and the peripheral system (the nerves). The adult brain weighs about 1300-1400 grams. Learn the brain part-to-function table; RPSC asks nothing else here.

Parts of the brain and what each controls

Part	Controls	Note
Cerebrum	Intelligence, memory, will, voluntary action, speech	The largest part of the brain
Cerebellum	Balance, posture and muscular coordination	Second largest; damage causes loss of balance
Medulla oblongata	Heartbeat, breathing, blood pressure, swallowing	The vital involuntary centre
Hypothalamus	Body temperature, hunger, thirst, sleep	The link between the nervous and endocrine systems
Thalamus	Relay station for sensory impulses	Passes signals up to the cerebrum
Pons	Bridge between brain parts; helps regulate breathing	Part of the hindbrain

- Humans have 12 pairs of cranial nerves and 31 pairs of spinal nerves.
- Reflex action is mediated by the SPINAL CORD through a reflex arc - the brain is informed afterwards.
- The neuron carries the impulse from dendrite to cell body to axon; the gap between two neurons is the synapse.
- Sympathetic nerves prepare the body for emergency; parasympathetic nerves calm it down.

- Eye - rods hold rhodopsin (visual purple) and work in dim light; cones give colour vision.
- The yellow spot or fovea has the maximum visual acuity; the blind spot has no photoreceptors at all.
- Ear ossicles, the three smallest bones - malleus, incus, stapes (hammer, anvil, stirrup).
- The cochlea is for hearing; the three semicircular canals are for balance.
- Tongue detects taste; the nose has olfactory receptors; skin is the largest sense organ.

TRAP

The commonest planted error - 'balance and coordination are controlled by the medulla oblongata'. No. Balance is the CEREBELLUM; the medulla runs heartbeat, breathing and blood pressure. Also, the cochlea is part of the ear, not the brain.

Endocrine glands, and the last sweep - bones, muscles, evolution

Endocrine glands are ductless - they pour hormones straight into the blood. The pituitary sits at the base of the brain and is called the master gland because it controls the others, though it is itself controlled by the hypothalamus. Learn the table in three columns - gland, hormone, and the disorder when it goes wrong. RPSC's favourite trick is a gland-hormone pair that does not match.

Endocrine glands, hormones and disorders

Gland	Hormone	Deficiency / excess
Pituitary (anterior)	Growth hormone	Deficiency - dwarfism; excess in childhood - gigantism; excess in an adult - acromegaly

Gland	Hormone	Deficiency / excess
Pituitary (posterior)	ADH / vasopressin	Deficiency - diabetes insipidus (large volumes of dilute urine)
Thyroid	Thyroxine (needs iodine)	Deficiency - goitre, cretinism in infants, myxoedema in adults; excess - Graves' disease
Thyroid	Calcitonin	Lowers blood calcium - note, calcitonin is thyroid, NOT parathyroid
Parathyroid	Parathormone	Raises blood calcium; deficiency causes tetany
Adrenal cortex	Cortisol and aldosterone	Deficiency - Addison's disease; excess - Cushing's syndrome
Adrenal medulla	Adrenaline and noradrenaline	The emergency or fight-or-flight hormone
Pancreas - beta cells of the islets of Langerhans	Insulin	Deficiency - diabetes mellitus (insulin lowers blood glucose)
Pancreas - alpha cells	Glucagon	Raises blood glucose - antagonistic to insulin
Testis	Testosterone	Male secondary sexual characters
Ovary	Oestrogen and progesterone	Female secondary characters; progesterone maintains pregnancy
Thymus	Thymosin	Maturation of T-lymphocytes; shrinks after puberty
Pineal	Melatonin	Sleep-wake rhythm

- Master gland = pituitary, lying in the sella turcica; its own master is the hypothalamus.
- Diabetes MELLITUS is insulin failure and sugar in the urine; diabetes INSIPIDUS is ADH failure and no sugar at all.
- Adrenaline is the hormone of fright, flight and fight - it raises heart rate, blood sugar and alertness.
- The thyroid needs iodine; that single link ties this chapter to goitre and to salt iodisation.

- An adult has 206 bones; a newborn has about 300, many of which fuse during growth.
- Skull has 22 bones; the femur (thigh) is the longest bone, the stapes of the middle ear the smallest.
- Gluteus maximus of the buttock is the largest muscle; the stapedius of the middle ear the smallest.
- Muscle types - skeletal (striated, voluntary), smooth (unstriated, involuntary), cardiac (striated, involuntary, never fatigues).
- Human evolution sequence - Australopithecus, then Homo habilis (the 'handy man', first tool maker), Homo erectus, Homo neanderthalensis, Homo sapiens.
- Humans carry 46 chromosomes in 23 pairs - 22 pairs of autosomes plus one pair of sex chromosomes, XX in the female and XY in the male.

CURRENT LINK

Current link, as of September 2026 - the Nobel Prize in Physiology or Medicine 2024 went to Victor Ambros and Gary Ruvkun for the discovery of microRNA and its role in gene regulation inside the cell. RAS has asked cell-biology Nobels before, so keep the latest year's medicine Nobel on your revision card.

REVISION RECAP

1. Hooke coined 'cell' (1665, cork); Brown found the nucleus (1831); Schleiden + Schwann gave cell theory, Virchow added 'omnis cellula e cellula'.
2. Mitochondrion = powerhouse, cristae, own circular DNA, 70S ribosomes. Lysosome = suicide bag (de Duve). Chloroplast = kitchen, magnesium in chlorophyll.
3. Ribosome 70S in prokaryotes and in mitochondria/chloroplasts; 80S in eukaryotic cytoplasm.
4. Plant cell only - cellulose wall, plastids, big central vacuole. Animal cell only - centriole, prominent lysosomes.
5. Mitosis = somatic, equational, 2 identical diploid cells. Meiosis = germ cells, reductional, 4 haploid cells, crossing over at pachytene.

6. Fat 9 kcal/g against 4 for carbohydrate and protein. Enzymes are protein (ribozymes excepted) and lower activation energy.
7. DNA - Watson and Crick 1953; A=T two bonds, G-C three; RNA has uracil and ribose.
8. Whittaker 1969 gave five kingdoms; Linnaeus gave binomial nomenclature, genus first. Photosynthesis - light reaction in grana, Calvin cycle in stroma, oxygen from splitting water.
9. Auxin apical dominance, gibberellin bolting, cytokinin cell division, ABA stress and stomatal closure, ethylene ripening.
10. Ptyalin on starch in the mouth, pepsin + HCl on protein in the stomach, trypsin from the pancreas; bile has no enzyme; absorption in the small intestine.
11. Liver is the largest gland - bile, urea, glycogen. Pancreas is mixed. Appendix is vestigial.
12. SA node in the RIGHT atrium is the pacemaker; lub = AV valves, dub = semilunar valves; 120/80 mm Hg; Harvey described circulation.
13. Plasma 55%; RBC 4.5-5.5 million/microlitre, 120 days, anucleate, destroyed in the spleen; platelets 1.5-4.5 lakh.
14. O = universal donor (no antigen, both antibodies); AB = universal recipient (both antigens, no antibody); Landsteiner 1901, Nobel 1930.
15. ABO is three alleles on chromosome 9 - I-A and I-B codominant, i recessive; O child = ii.
16. Rh named after the Rhesus monkey, Landsteiner and Wiener 1940; Rh-negative mother + Rh-positive foetus risks erythroblastosis foetalis from the second pregnancy; anti-D prevents it.
17. Nephron is the kidney's unit (10 lakh each); Malpighian corpuscle = Bowman's capsule + glomerulus; humans are ureotelic; ADH saves water.
18. Cerebrum largest, cerebellum balance, medulla heartbeat and breathing, hypothalamus temperature and thirst; 12 cranial + 31 spinal nerves; reflex through the spinal cord.
19. Pituitary is the master gland - GH gives dwarfism/gigantism/acromegaly; thyroxine needs iodine; insulin from beta cells; adrenaline is the emergency hormone.
20. 206 bones, femur longest, stapes smallest, gluteus maximus largest muscle; 46 chromosomes; Homo habilis was the first tool maker.